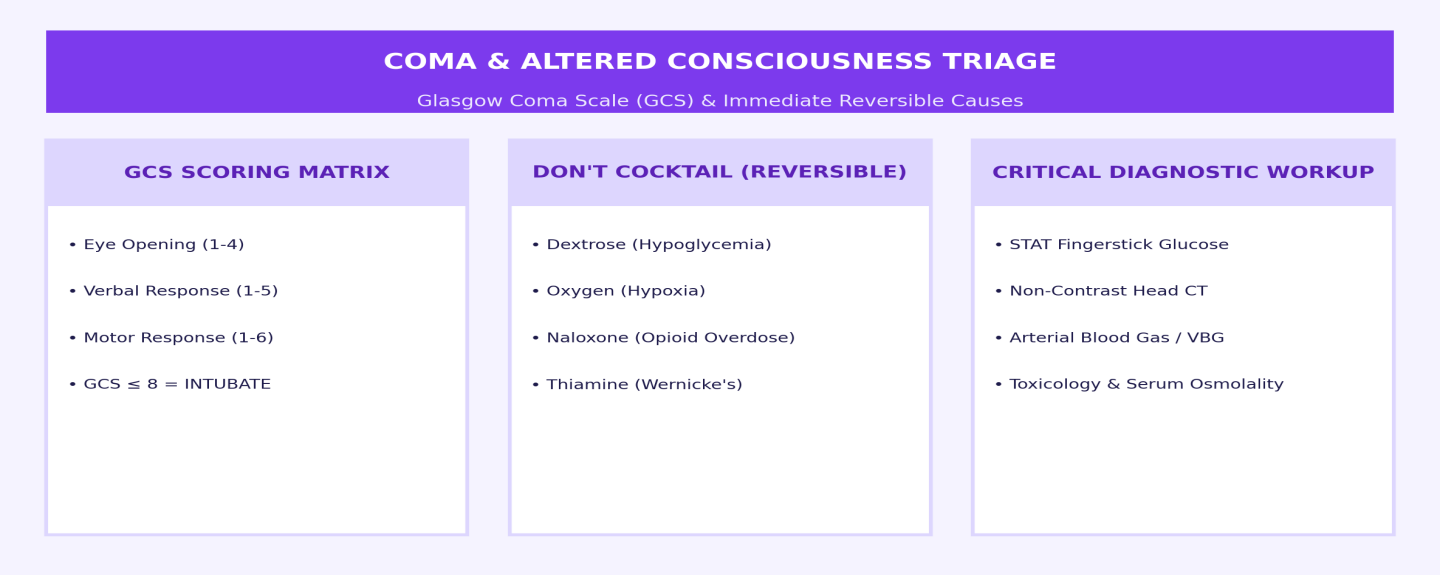
EMERGENCY MEDICINE ROTATION REVIEW

Chapter 5: Neurology System — High-Yield Board Review Guide

1. Altered Level of Consciousness & Coma

Pathophysiology & Etiology: Coma results from bilateral cerebral hemisphere dysfunction or acute Reticular Activating System (RAS) disruption. Causes include structural lesions (trauma, stroke, bleed) and toxic-metabolic encephalopathy (hypoglycemia, hypoxia, sepsis, overdoses).
Clinical Presentation: Depressed sensorium, unresponsiveness, abnormal breathing patterns (Cheyne-Stokes), and loss of protective airway reflexes.
Physical Examination: Assess Glasgow Coma Scale (GCS 3-15), pupil reactivity (fixed dilated = herniation risk), doll's eyes (oculocephalic), and motor response (decorticate vs decerebrate posturing).
Diagnostic Studies: STAT fingerstick glucose, Non-Contrast Head CT, ABG/VBG, serum osmolality, toxicology screen, CMP, and LP if CNS infection suspected.
Management & Treatment: Airway protection (Intubate if GCS ≤ 8). Empirical 'DON'T' cocktail: Dextrose (50% IV), Oxygen, Naloxone (0.4-2mg IV), and Thiamine (100mg IV prior to glucose in suspected Wernicke's).

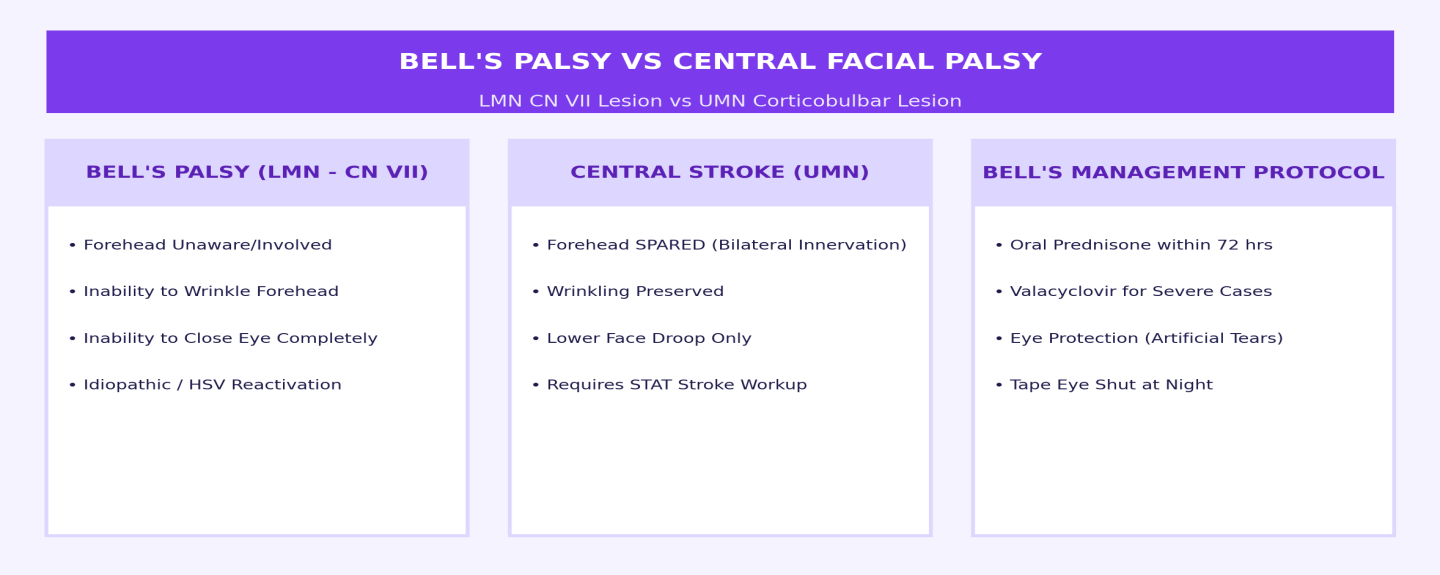


BOARD PEARL: Altered Level of Consciousness & Coma

GCS ≤ 8 mandates immediate endotracheal intubation for airway protection. Always administer Thiamine BEFORE Glucose in malnourished or alcoholic patients to prevent acute Wernicke encephalopathy precipitation.

2. Bell's Palsy vs Central Facial Palsy

Pathophysiology & Etiology: Bell's palsy is an acute, idiopathic lower motor neuron (LMN) paresis of Cranial Nerve VII, strongly linked to Herpes Simplex Virus (HSV-1) reactivation and nerve inflammation/edema within the facial canal.
Clinical Presentation: Sudden onset unilateral facial droop, inability to close eye, loss of taste on anterior 2/3 of tongue, hyperacusis (stapedius paralysis), and dry eye/mouth.
Physical Examination: LMN lesion affects BOTH upper and lower face (inability to wrinkle forehead or raise eyebrow). Central UMN stroke SPARES the forehead due to bilateral cortical innervation.
Diagnostic Studies: Clinical diagnosis. Non-contrast Head CT or MRI indicated if forehead is spared or atypical features are present.
Management & Treatment: Oral Prednisone (60mg daily x 5-7 days) initiated within 72 hours. Add Valacyclovir (1000mg TID) for severe cases. Crucial eye protection: artificial tears during day, lubricating ointment & taping eye shut at night.

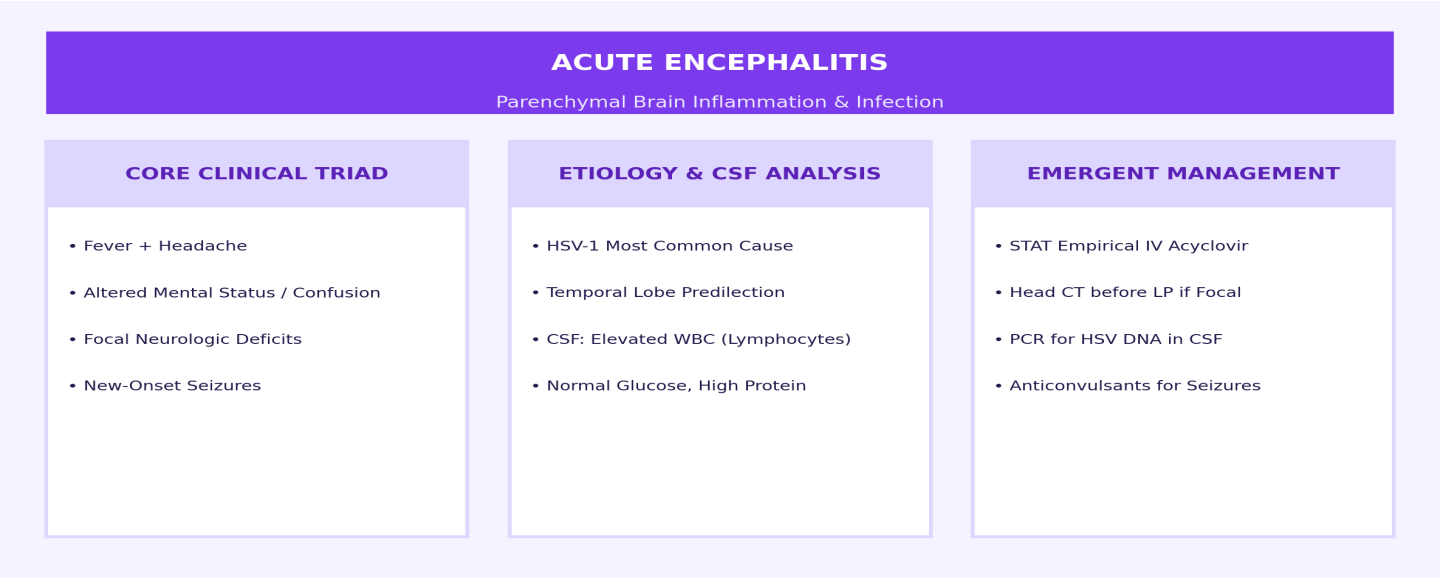


BOARD PEARL: Bell's Palsy vs Central Facial Palsy

The key clinical differentiator: Bell's palsy (LMN) involves the entire half of the face including forehead wrinkling. If forehead wrinkling is SPARED, think UMN Central Stroke and order a STAT Head CT!

3. Acute Encephalitis

Pathophysiology & Etiology: Acute infection and parenchymal inflammation of the brain tissue. Herpes Simplex Virus 1 (HSV-1) is the most common and lethal viral etiology, targeting the temporal lobes. Other causes: West Nile, VZV, EBV, enteroviruses.
Clinical Presentation: Encephalitis triad: Fever, Headache, and Altered Mental Status / Personality Changes. Accompanied by focal neurologic deficits, behavioral changes, or new-onset seizures.
Physical Examination: Confusion, disorientation, dysphasia, memory impairment, cranial nerve palsies, hyperreflexia, and meningeal signs if meningoencephalitis.
Diagnostic Studies: Non-contrast Head CT (temporal lobe hypodensity), Lumbar Puncture (CSF: lymphocytic pleocytosis, RBCs, normal glucose, elevated protein), CSF HSV PCR (gold standard), MRI (temporal lobe signal hyperintensity).
Management & Treatment: Emergent IV Acyclovir (10mg/kg IV q8h) started immediately upon suspicion before LP/PCR confirmation. Anticonvulsants for seizures, ICP monitoring if severe.

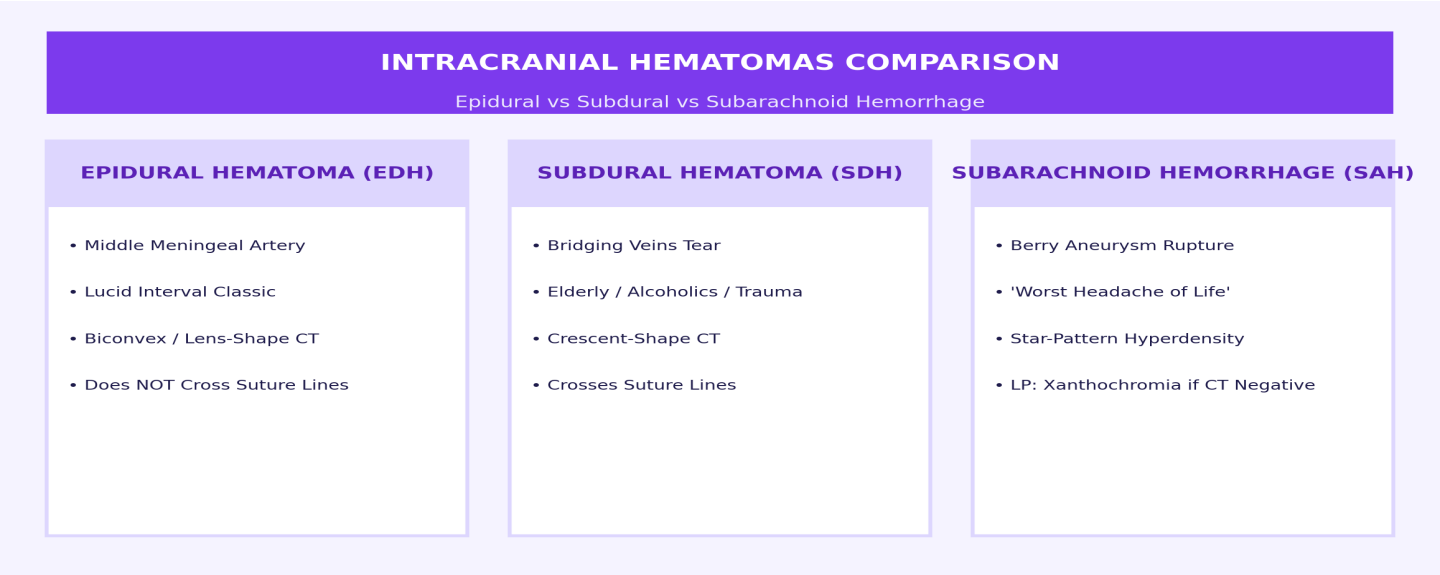


BOARD PEARL: Acute Encephalitis

Initiate empirical IV Acyclovir immediately in any patient presenting with fever, altered mental status, and psychiatric or temporal lobe symptoms—do not wait for CSF PCR results!

4. Intracranial Hematomas Comparison

Pathophysiology & Etiology: Epidural Hematoma (EDH): Arterial bleed (Middle Meningeal Artery) secondary to temporoparietal skull fracture. Subdural Hematoma (SDH): Venous bleed from torn bridging veins (elderly, alcoholics, infants). Subarachnoid Hemorrhage (SAH): Ruptured berry aneurysm.
Clinical Presentation: EDH: Classic 'Lucid Interval' (brief loss of consciousness → transient recovery → rapid deterioration). SDH: Insidious headache, confusion, progressive focal deficits. SAH: Sudden 'thunderclap' headache.
Physical Examination: EDH: Ipsilateral dilated pupil (CN III compression) and contralateral hemiparesis. SDH: Fluctuating GCS. SAH: Nuchal rigidity, Kernig/Brudzinski signs.
Diagnostic Studies: Non-Contrast Head CT. EDH shows biconvex (lens-shaped) hyperdensity NOT crossing suture lines. SDH shows crescent-shaped hyperdensity crossing suture lines. SAH shows hyperdensity in basal cisterns.
Management & Treatment: Emergent neurosurgical evacuation for EDH/SDH with mass effect or midline shift > 5mm. Reversal of anticoagulation, ICP control (Mannitol, hypertonic saline).

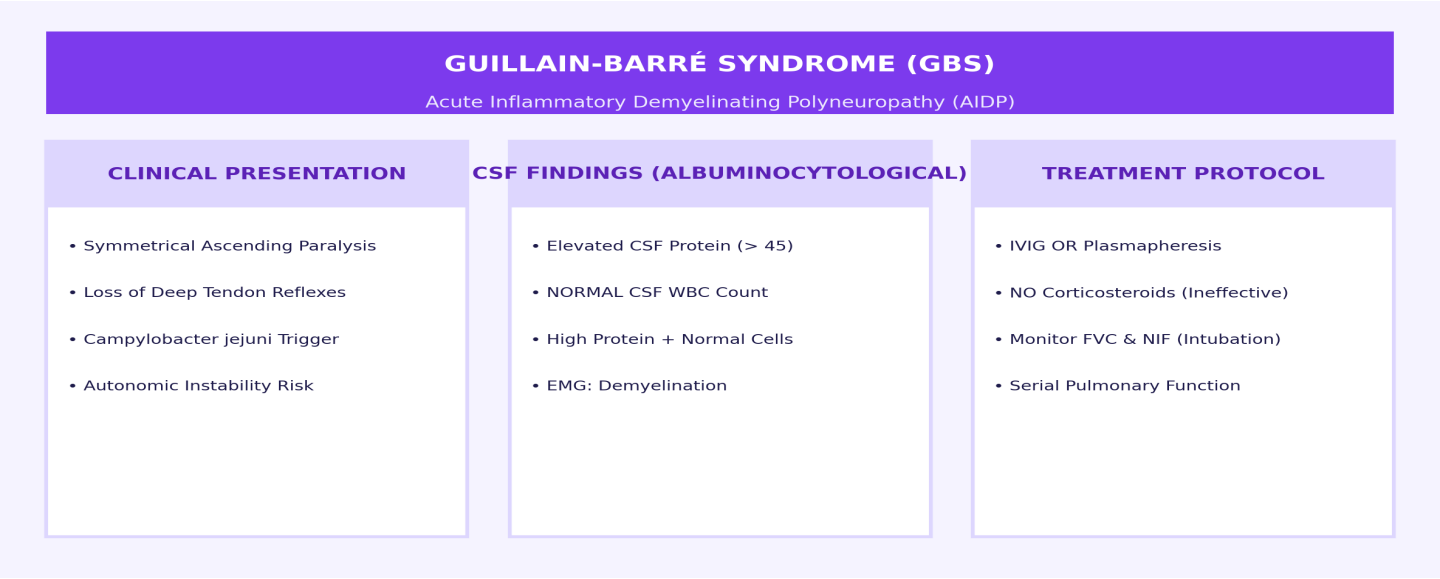


BOARD PEARL: Intracranial Hematomas Comparison

Epidural hematoma = Lens-shaped / Biconvex, does NOT cross suture lines (Middle Meningeal Artery). Subdural hematoma = Crescent-shaped, DOES cross suture lines (Bridging Veins).

5. Guillain-Barré Syndrome (GBS)

Pathophysiology & Etiology: Acute Inflammatory Demyelinating Polyneuropathy (AIDP) triggered by molecular mimicry following gastrointestinal or respiratory infection, most commonly *Campylobacter jejuni*, CMV, or EBV.
Clinical Presentation: Symmetric, ascending muscle weakness starting in lower extremities and spreading proximally. Paresthesias, neuropathic pain, and autonomic instability (arrhythmias, labile BP).
Physical Examination: Areflexia (loss of Deep Tendon Reflexes) in affected limbs. Cranial nerve involvement (facial diplegia in 50%). Respiratory muscle weakness.
Diagnostic Studies: Lumbar Puncture showing *Albuminocytological Dissociation* (elevated CSF protein with NORMAL WBC count). Serial Forced Vital Capacity (FVC) and Negative Inspiratory Force (NIF) monitoring.
Management & Treatment: Intravenous Immunoglobulin (IVIG 0.4g/kg/day x 5 days) OR Plasmapheresis. Steroids are INEFFECTIVE. Intubate if FVC < 20 mL/kg or NIF < -30 cmH2O.

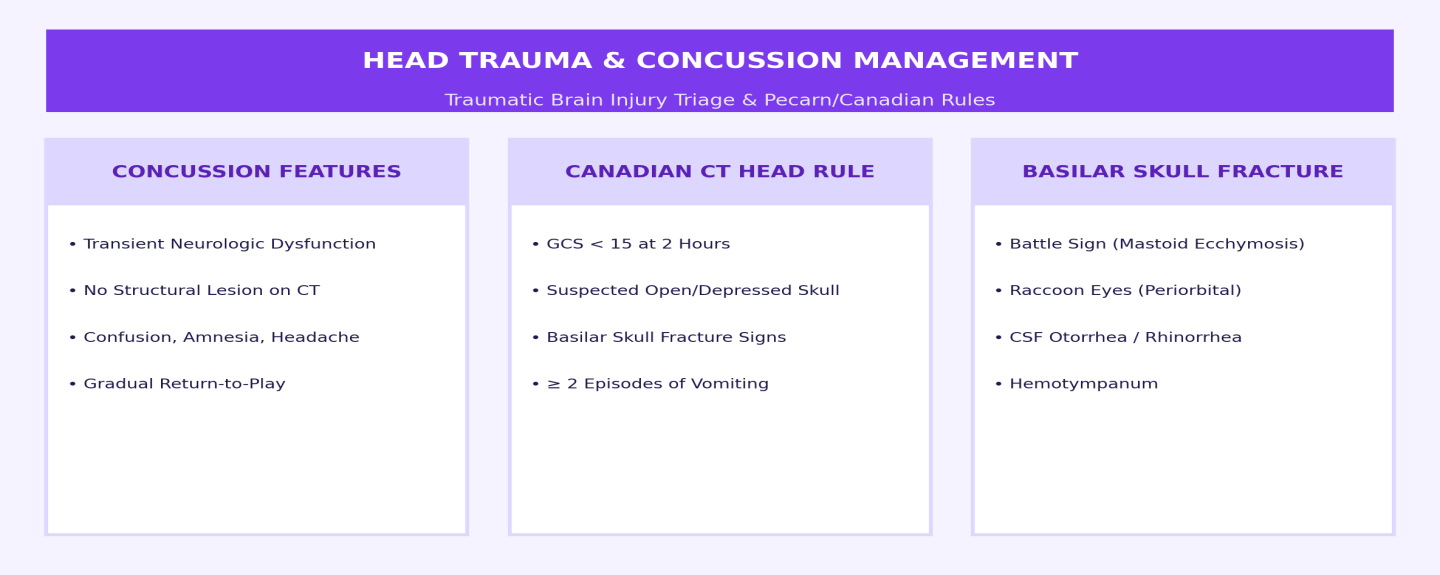


BOARD PEARL: Guillain-Barré Syndrome (GBS)

Classic CSF finding in GBS is Albuminocytological Dissociation (High CSF Protein with Normal Cell Count). Monitor FVC/NIF closely; intubation is needed if FVC drops below 20 mL/kg.

6. Head Trauma & Concussion Management

Pathophysiology & Etiology: Traumatic Brain Injury (TBI) ranging from mild concussion (transient neuronal dysfunction without structural lesion) to severe contusions and diffuse axonal injury (DAI).
Clinical Presentation: Confusion, retrograde/anterograde amnesia, headache, dizziness, nausea, photophobia, and transient loss of consciousness.
Physical Examination: Assess GCS, focal neurologic deficits, signs of basilar skull fracture (Battle's sign, Raccoon eyes, hemotympanum, CSF otorrhea/rhinorrhea).
Diagnostic Studies: Apply Canadian CT Head Rule or PECARN (pediatrics) to guide Non-Contrast Head CT usage. CT indicated if GCS < 15 at 2h, suspected open skull fracture, vomiting ≥ 2 times, or age ≥ 65.
Management & Treatment: Cognitive and physical rest. Graduated step-wise return-to-play protocol. Strict avoidance of contact sports during recovery to prevent Second Impact Syndrome.

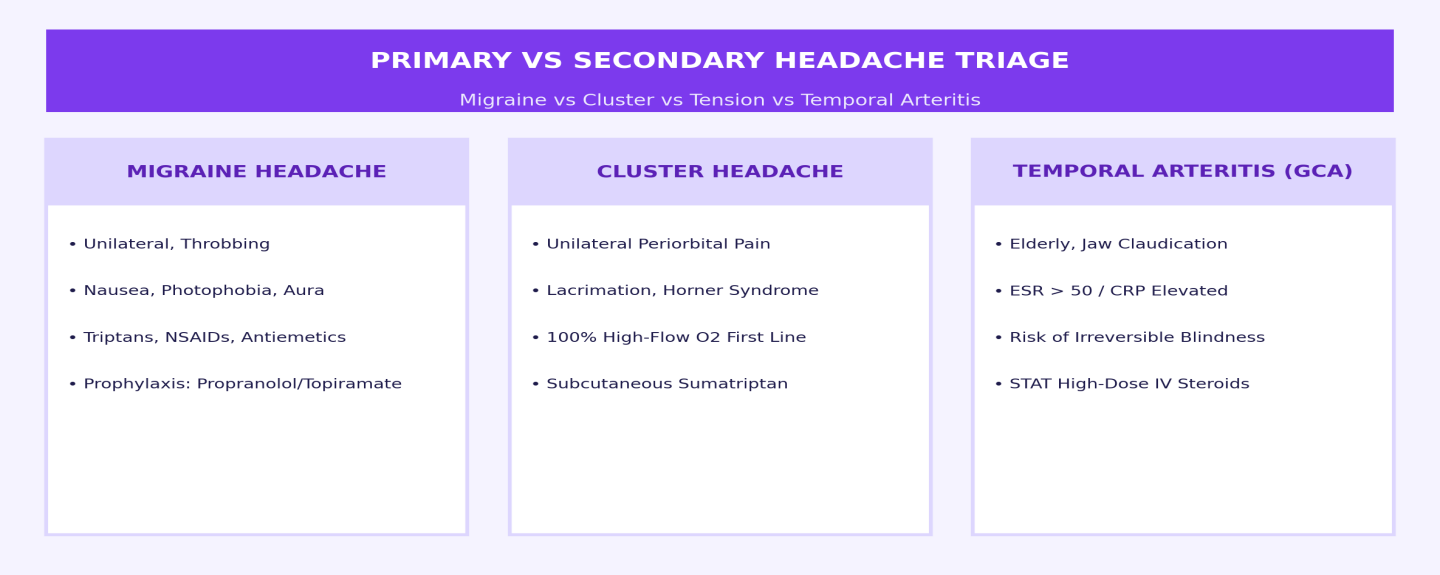


BOARD PEARL: Head Trauma & Concussion Management

Basilar skull fracture signs: Battle sign (mastoid ecchymosis), Raccoon eyes (periorbital ecchymosis), hemotympanum, and CSF rhinorrhea/otorrhea. Avoid nasogastric tubes in these patients!

7. Primary vs Secondary Headache Triage

Pathophysiology & Etiology: Migraine (neurovascular inflammation/CGRP release), Cluster (trigeminal autonomic cephalgia), Tension (pericranial muscle tightness), Temporal Arteritis (systemic granulomatous vasculitis of medium/large arteries).
Clinical Presentation: Migraine: Unilateral throbbing w/ nausea, photophobia, aura. Cluster: Severe unilateral periorbital/temporal piercing pain. Temporal Arteritis: Unilateral headache, jaw claudication, visual changes in elderly.
Physical Examination: Cluster: Ipsilateral Horner syndrome (ptosis, miosis), lacrimation, rhinorrhea. Temporal Arteritis: Tender, nodular, non-pulsatile temporal artery.
Diagnostic Studies: Temporal Arteritis: ESR > 50 mm/hr, CRP elevated, temporal artery biopsy. CT/MRI for red flags (SNOOP: Systemic signs, Neurologic deficits, Onset sudden, Older age, Pattern change).
Management & Treatment: Cluster: 100% High-Flow Oxygen (12-15 L/min NRB) + Subcutaneous Sumatriptan. Temporal Arteritis: Immediate High-Dose IV Steroids (Methylprednisolone) to prevent irreversible blindness.

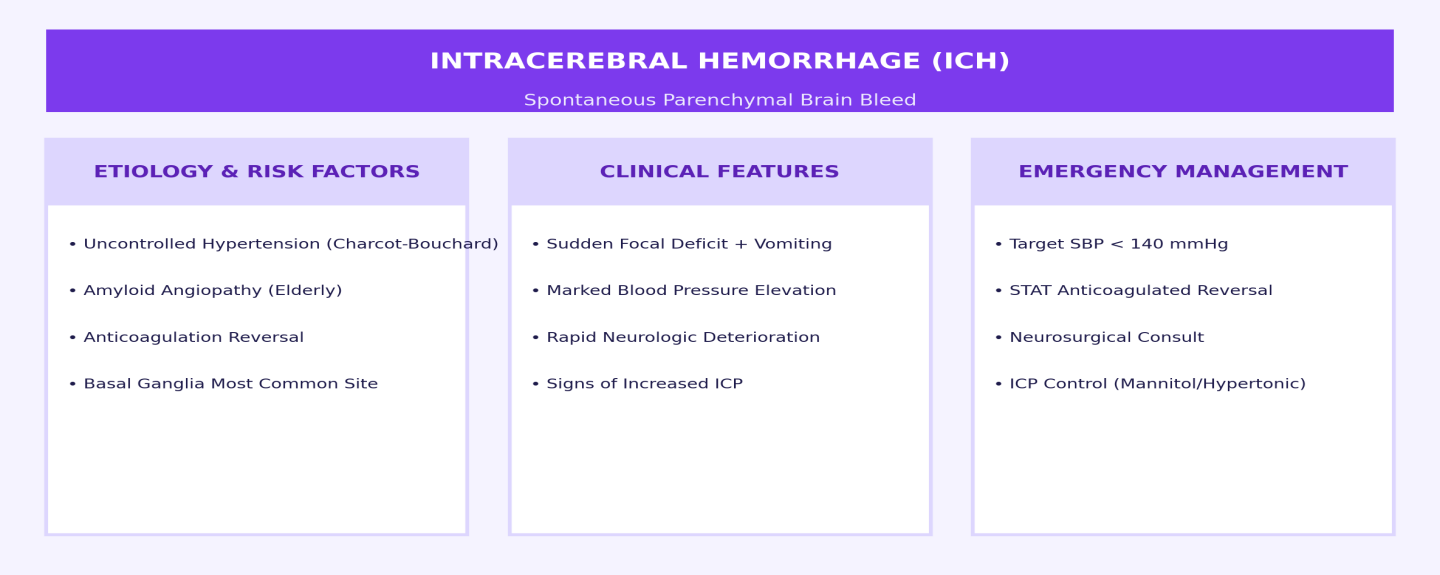


BOARD PEARL: Primary vs Secondary Headache Triage

In suspected Temporal (Giant Cell) Arteritis, initiate High-Dose Systemic Corticosteroids IMMEDIATELY before biopsy to prevent permanent visual loss!

8. Intracerebral Hemorrhage (ICH)

Pathophysiology & Etiology: Spontaneous bleeding into brain parenchyma. Chronic uncontrolled hypertension is the main cause (rupture of Charcot-Bouchard microaneurysms in basal ganglia, thalamus, pons). Cerebral Amyloid Angiopathy in elderly.
Clinical Presentation: Sudden onset focal neurologic deficit rapidly progressing over minutes to hours, accompanied by severe headache, nausea, vomiting, and altered mental status.
Physical Examination: Severe hypertension (BP > 180/110), focal weakness/sensory loss, Cushing's Triad if elevated ICP (hypertension, bradycardia, irregular respirations).
Diagnostic Studies: STAT Non-Contrast Head CT (hyperdense intraparenchymal mass w/ surrounding edema). CTA/MRA to rule out AVM or aneurysm. Coagulation profile (PT/INR, PTT, Platelets).
Management & Treatment: Rapid BP reduction (Target SBP 130-140 mmHg using IV Labetalol or Nicardipine). Immediate reversal of anticoagulation (PCC, Vitamin K, Idarucizumab). Neurosurgical consult.

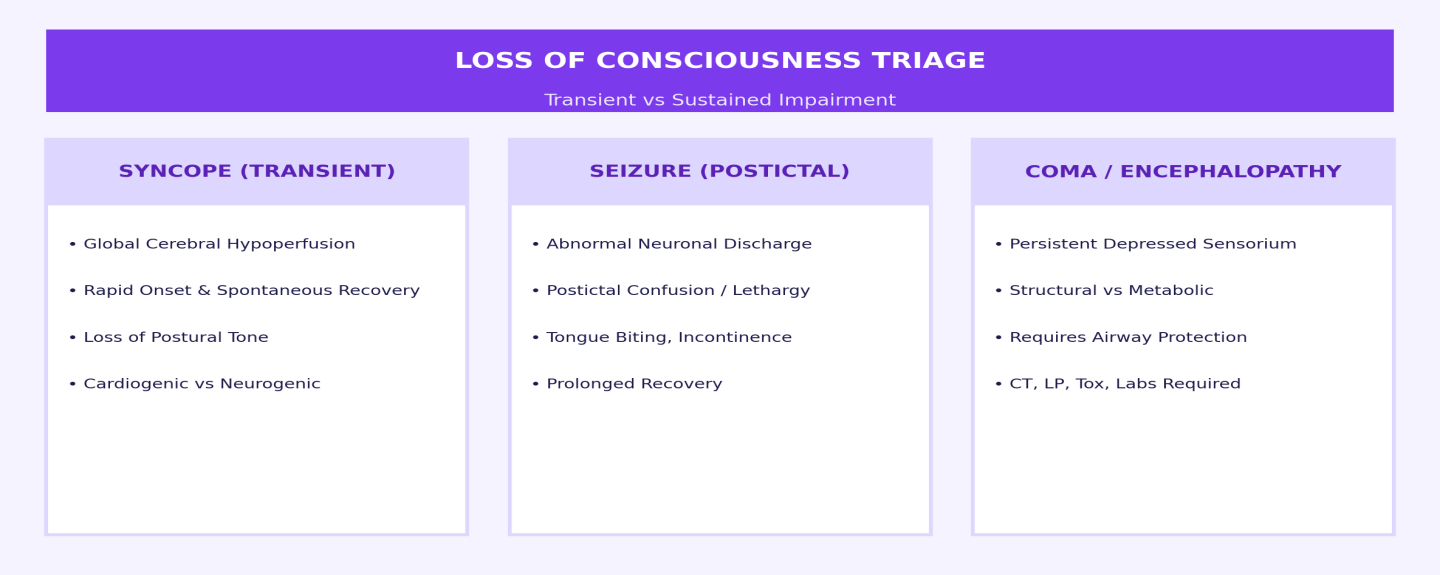


BOARD PEARL: Intracerebral Hemorrhage (ICH)

Basal ganglia (putamen) is the most common site for hypertensive ICH. Target rapid blood pressure reduction (SBP < 140 mmHg) and reverse any active anticoagulants STAT!

9. Loss of Consciousness Triage

Pathophysiology & Etiology: Transient loss of consciousness (TLOC) differentiates into Syncope (global self-resolving cerebral hypoperfusion), Seizures (abnormal cortical hypersynchrony), and Coma/Encephalopathy (sustained cerebral/RAS dysfunction).
Clinical Presentation: Syncope: Brief loss of postural tone, rapid complete recovery, no postictal state. Seizure: Aura, tonic-clonic motor activity, tongue biting, incontinence, postictal confusion.
Physical Examination: Check orthostatic vital signs, cardiac auscultation (murmurs), trauma examination (tongue lacerations), detailed neurologic assessment.
Diagnostic Studies: STAT Fingerstick Glucose, 12-Lead EKG (essential for all syncope/seizure patients to rule out arrhythmia), CT Head if focal deficits or trauma.
Management & Treatment: Syncope: Treat underlying cause (cardiac workup if exertional/supine). Seizure: Ensure airway patency, check electrolytes/glucose, administer IV Benzodiazepines if ongoing.

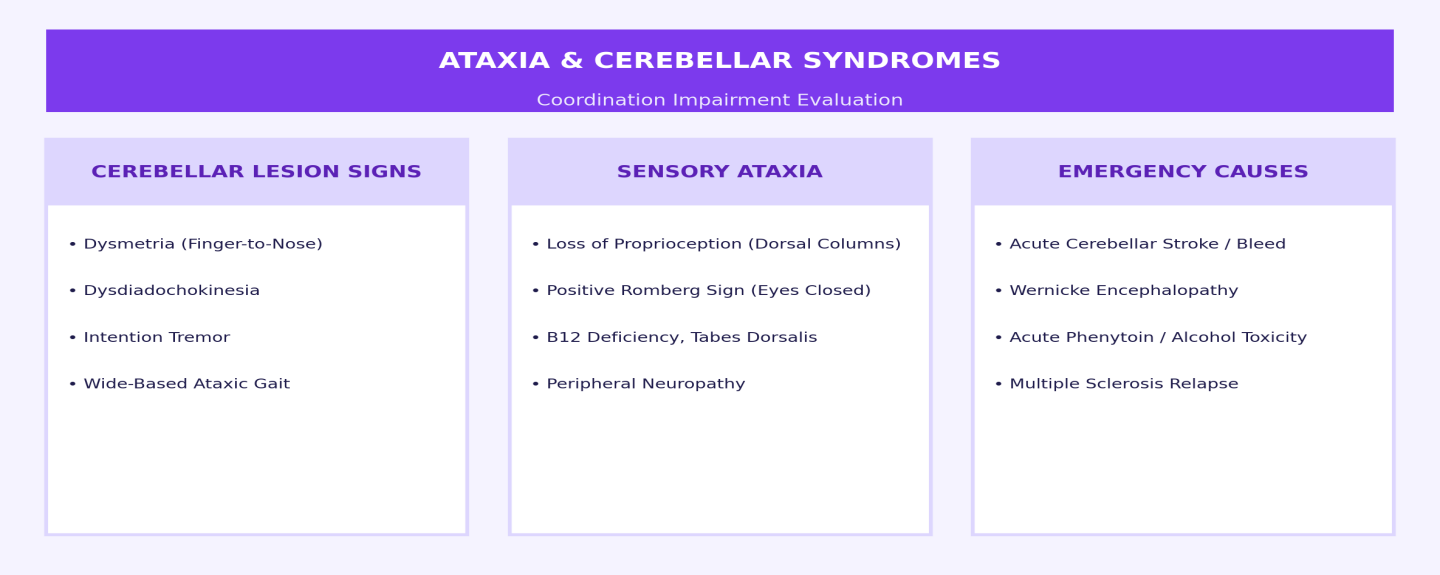


BOARD PEARL: Loss of Consciousness Triage

A 12-lead EKG is MANDATORY in all patients presenting with loss of consciousness to rule out life-threatening cardiac arrhythmias (Brugada, Long QT, WPW, Complete Block).

10. Ataxia & Cerebellar Syndromes

Pathophysiology & Etiology: Incoordination of muscle movements caused by Cerebellar lesions (stroke, bleed, tumor, MS, alcohol/phenytoin toxicity) or Sensory loss (dorsal column dysfunction, B12 deficiency, tabes dorsalis).
Clinical Presentation: Wide-based unsteady gait, intention tremor, frequent falls, slurred speech (scanning dysarthria), vertigo, and dizziness.
Physical Examination: Cerebellar: Dysmetria (past-pointing on finger-to-nose), dysdiadochokinesia (impaired rapid alternating movements), nystagmus. Sensory: Positive Romberg Sign (swaying w/ eyes closed).
Diagnostic Studies: STAT Non-Contrast Head CT/MRI to rule out cerebellar infarction or hemorrhage (can cause brainstem compression/herniation). Vitamin B12 levels, drug levels.
Management & Treatment: Acute cerebellar stroke/bleed requires urgent neurosurgical evaluation for posterior fossa decompression. Thiamine for Wernicke's.

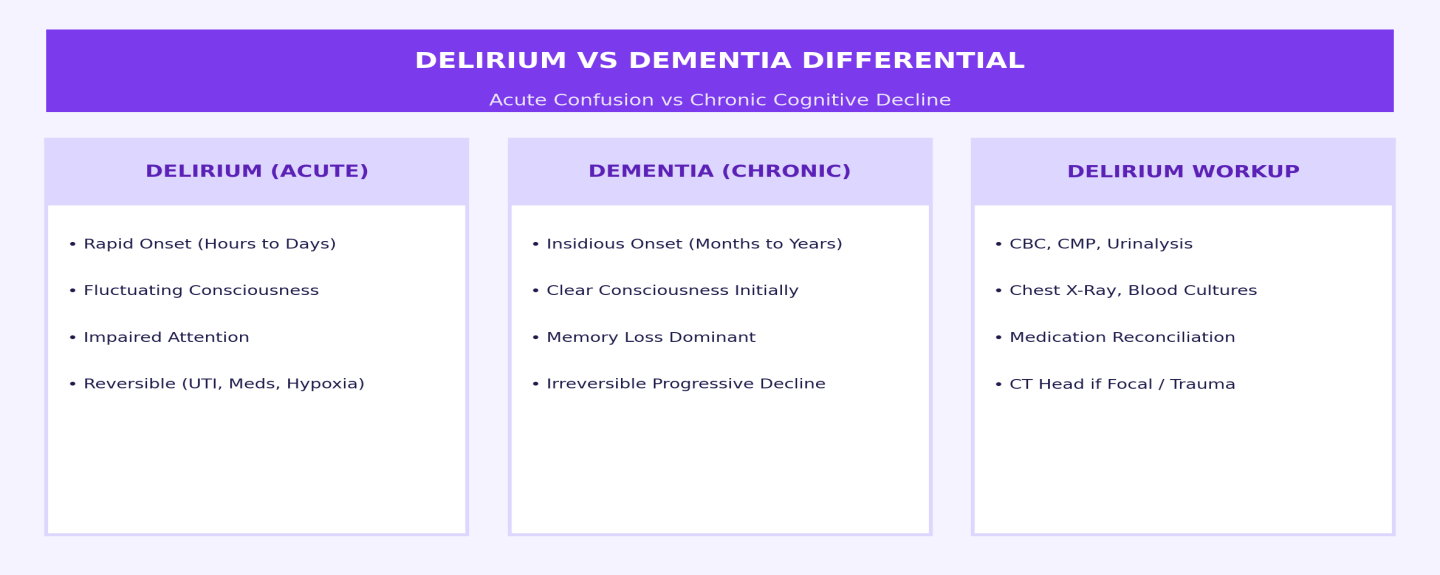


BOARD PEARL: Ataxia & Cerebellar Syndromes

Acute cerebellar stroke or hemorrhage is a surgical emergency! The rigid posterior fossa leaves no room for expansion; compression of the brainstem can cause rapid fatality.

11. Delirium vs Dementia Differential

Pathophysiology & Etiology: Delirium: Acute, reversible global cognitive impairment caused by medical illness (UTI, pneumonia, electrolyte imbalance, hypoxia, drug toxicity/withdrawal). Dementia: Chronic, progressive, irreversible neurodegeneration.
Clinical Presentation: Delirium: Sudden onset (hours/days), fluctuating course, marked inattention, altered level of consciousness, visual hallucinations. Dementia: Insidious onset (months/years), intact attention early.
Physical Examination: Confusion Assessment Method (CAM) criteria for delirium. Search for occult infection, distended bladder, fecal impaction.
Diagnostic Studies: CBC, CMP, Urinalysis & Urine Culture, CXR, ABG, EKG, Med reconciliation. CT Head if trauma, focal deficit, or new acute onset.
Management & Treatment: Identify and treat underlying cause. Non-pharmacologic reorientation. Avoid physical restraints and sedatives; Low-dose Haloperidol if severe agitation.

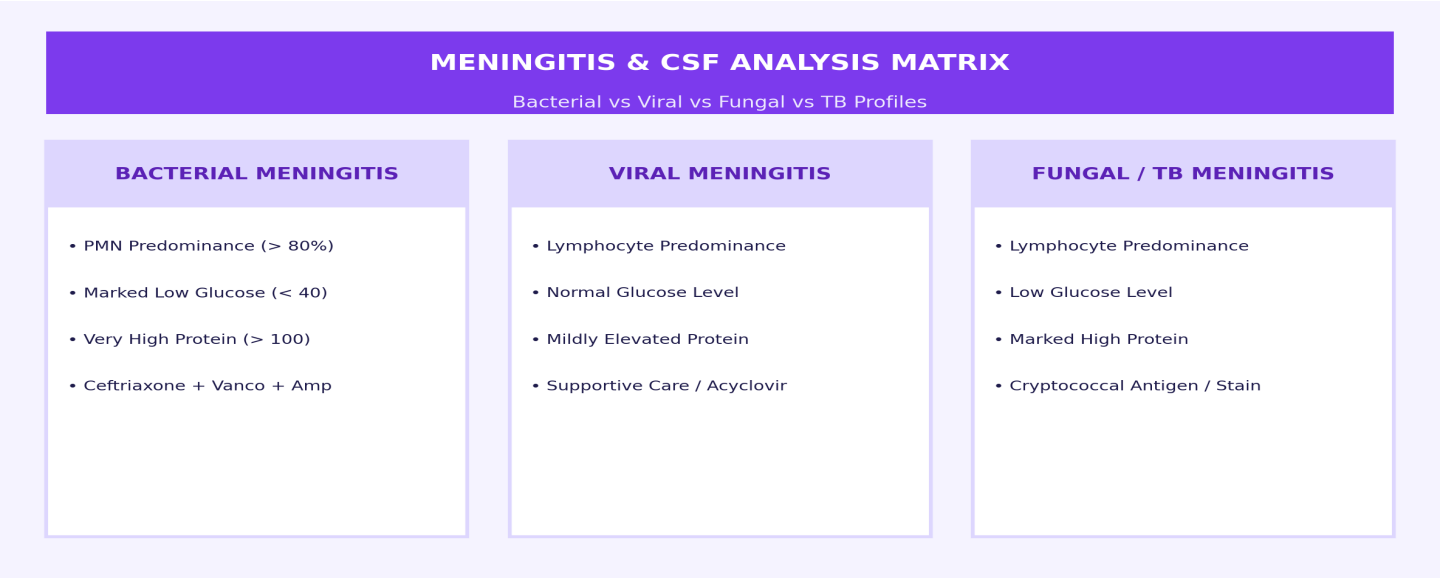


BOARD PEARL: Delirium vs Dementia Differential

Delirium is an ACUTE MEDICAL EMERGENCY characterized by fluctuating consciousness and impaired attention. Always search for hidden infections (UTI, Pneumonia) or drug toxicity!

12. Meningitis & CSF Analysis Matrix

Pathophysiology & Etiology: Infection of the leptomeninges. Bacterial: *S. pneumoniae*, *N. meningitidis*, *L. monocytogenes* (elderly/infants/immunosuppressed). Viral: Enteroviruses, HSV. Fungal: *Cryptococcus*.
Clinical Presentation: Classic Triad: Fever, Nuchal Rigidity (stiff neck), and Altered Mental Status. Photophobia, severe headache, petechial rash (N. meningitidis).
Physical Examination: Positive Kernig Sign (resistance to knee extension) & Brudzinski Sign (involuntary hip flexion on neck flexion). Jolt accentuation of headache.
Diagnostic Studies: Lumbar Puncture CSF Analysis: Bacterial (WBC > 1000 PMNs, Glucose < 40, Protein > 100). Viral (WBC < 500 Lymphs, Normal Glucose). Head CT before LP ONLY IF focal deficits, papilledema, or GCS < 10.
Management & Treatment: Empiric IV Ceftriaxone (2g) + Vancomycin (15-20mg/kg) + Ampicillin (if > 50 yo). Administer IV Dexamethasone PRIOR to or with first dose of antibiotics.

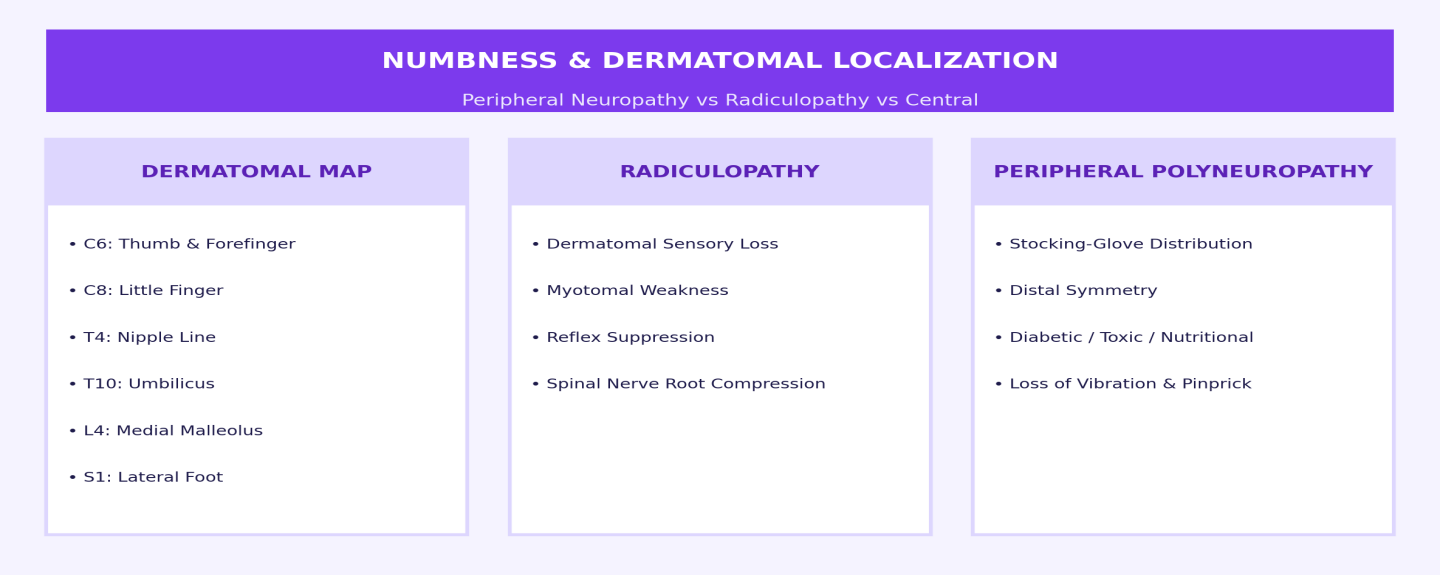


BOARD PEARL: Meningitis & CSF Analysis Matrix

Do NOT delay empiric antibiotics for a Head CT! If CT is required before LP, draw blood cultures and start IV Ceftriaxone + Vancomycin + Dexamethasone IMMEDIATELY.

13. Numbness & Dermatomal Localization

Pathophysiology & Etiology: Sensory deficits localize along central pathways, spinal nerve roots (radiculopathy), plexus, or peripheral nerves (polyneuropathy). Common causes: herniated disc, diabetes, B12 deficiency, stroke.
Clinical Presentation: Radiculopathy: Sharp radiating pain and numbness along dermatomal distribution. Polyneuropathy: Distal symmetric 'stocking-glove' numbness.
Physical Examination: Key Dermatomes: C6 (Thumb), C8 (Little finger), T4 (Nipples), T10 (Umbilicus), L4 (Medial malleolus), L5 (Great toe / Dorsum foot), S1 (Lateral foot / Heel).
Diagnostic Studies: MRI of spine for suspected nerve root/cord compression. Electromyography (EMG) and Nerve Conduction Studies (NCS). HbA1c, B12, TSH.
Management & Treatment: Neuropathic pain management (Gabapentin, Pregabalin, Duloxetine). Physical therapy. Urgent surgical decompression for cauda equina or progressive motor deficits.

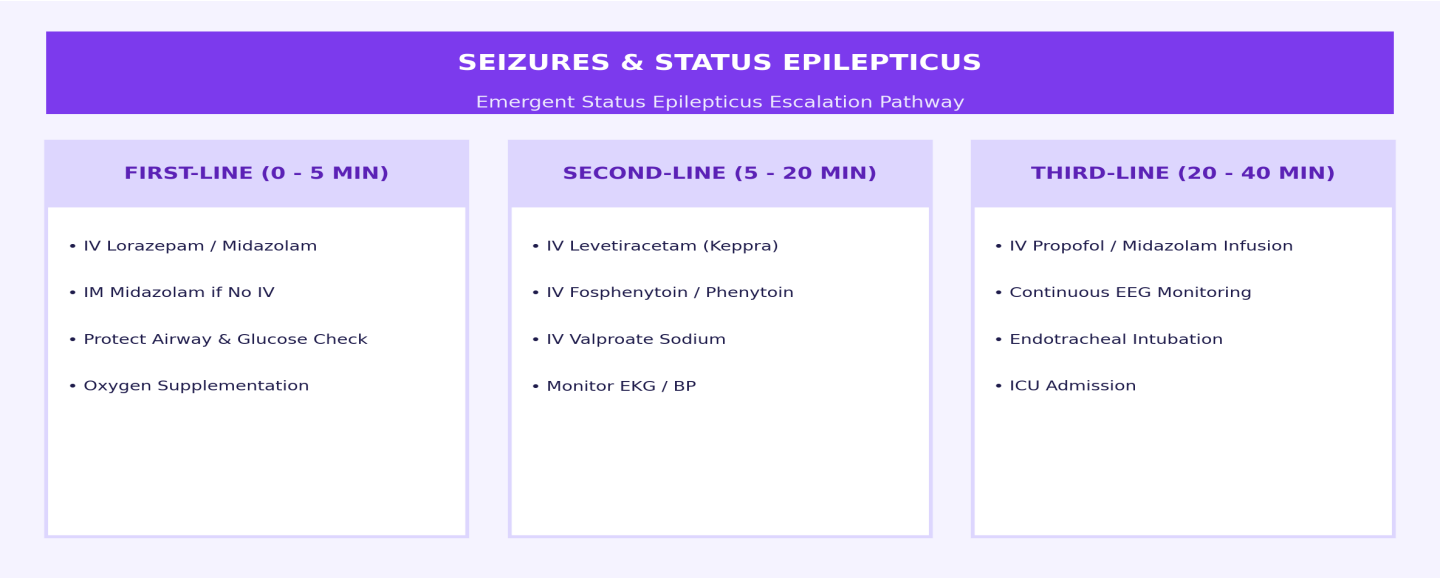


BOARD PEARL: Numbness & Dermatomal Localization

Know key dermatomes for rapid localization: C6 = Thumb, C8 = Pinky, T4 = Nipple line, T10 = Umbilicus, L4 = Medial Malleolus, L5 = Dorsum of Foot, S1 = Lateral Foot.

14. Seizures & Status Epilepticus

Pathophysiology & Etiology: Status Epilepticus: Continuous seizure activity lasting > 5 minutes or ≥ 2 discrete seizures without complete recovery of consciousness in between. Medical emergency causing permanent neuronal injury.
Clinical Presentation: Generalized tonic-clonic movements, unresponsive, cyanosis, autonomic hyperactivation, followed by prolonged postictal state.
Physical Examination: ABCs, airway patency, lateral tongue biting, pupil dilation, hyperreflexia, postictal Todd's paralysis (transient focal deficit).
Diagnostic Studies: Fingerstick glucose immediately. Electrolytes (Na, Ca, Mg), AED levels, Toxicology, ABG, Head CT after stabilization, EEG.
Management & Treatment: 0-5 min: IV Lorazepam (4mg) or Midazolam (10mg IM). 5-20 min: IV Levetiracetam (60mg/kg), Fosphenytoin (20mg PE/kg), or Valproate (40mg/kg). >20 min: IV Propofol/Midazolam infusion + Intubation.

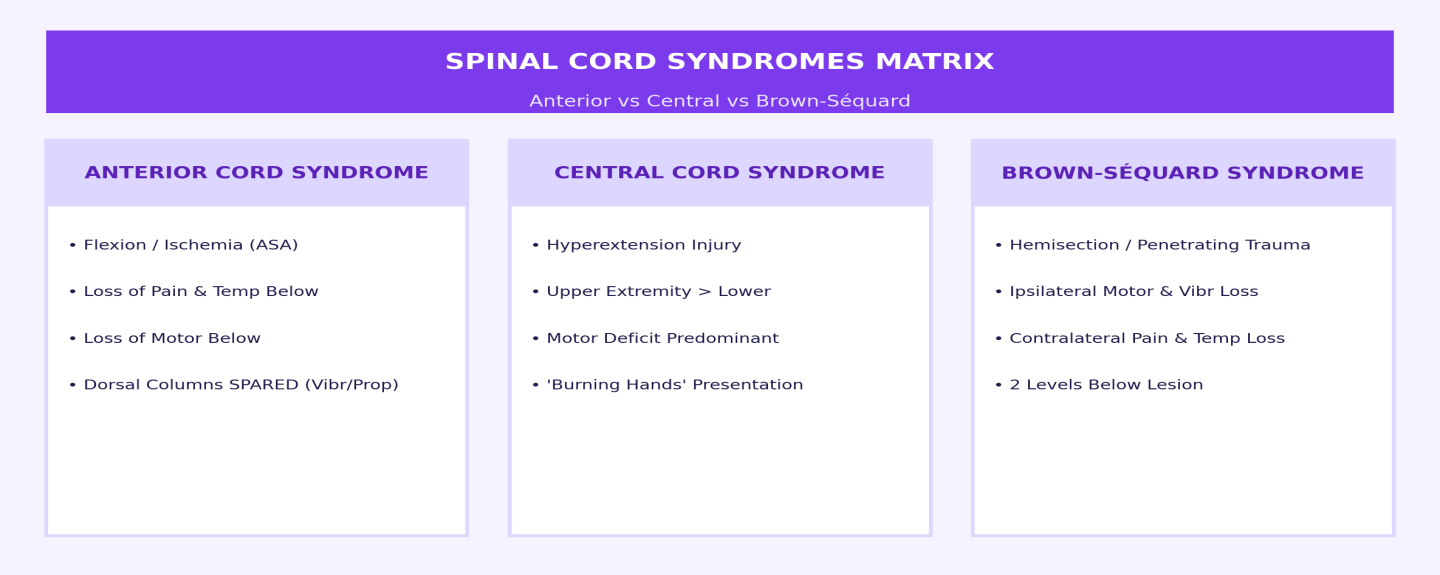


BOARD PEARL: Seizures & Status Epilepticus

Status Epilepticus First-Line: IV Lorazepam or IM Midazolam! If seizures persist > 5 min, escalate quickly to IV Levetiracetam or Fosphenytoin. Protect airway!

15. Spinal Cord Injuries & Syndromes

Pathophysiology & Etiology: Traumatic or ischemic cord injury. Anterior Cord: Flexion injury / ASA occlusion. Central Cord: Hyperextension injury in elderly. Brown-Séquard: Penetrating trauma (hemisection).
Clinical Presentation: Anterior Cord: Bilateral loss of motor function, pain, and temperature BELOW lesion; Dorsal Columns SPARED (vibration/proprioception intact). Central Cord: Motor loss Upper > Lower extremities ('burning hands').
Physical Examination: Brown-Séquard: Ipsilateral loss of motor & vibration/proprioception; Contralateral loss of pain/temp 1-2 levels below lesion.
Diagnostic Studies: STAT MRI Spine (gold standard for cord injury). CT Spine for bony fractures.
Management & Treatment: Rigid spinal immobilization (C-collar, log roll). Maintain MAP ≥ 85-90 mmHg (Norepinephrine) for cord perfusion. Neurosurgical consult.

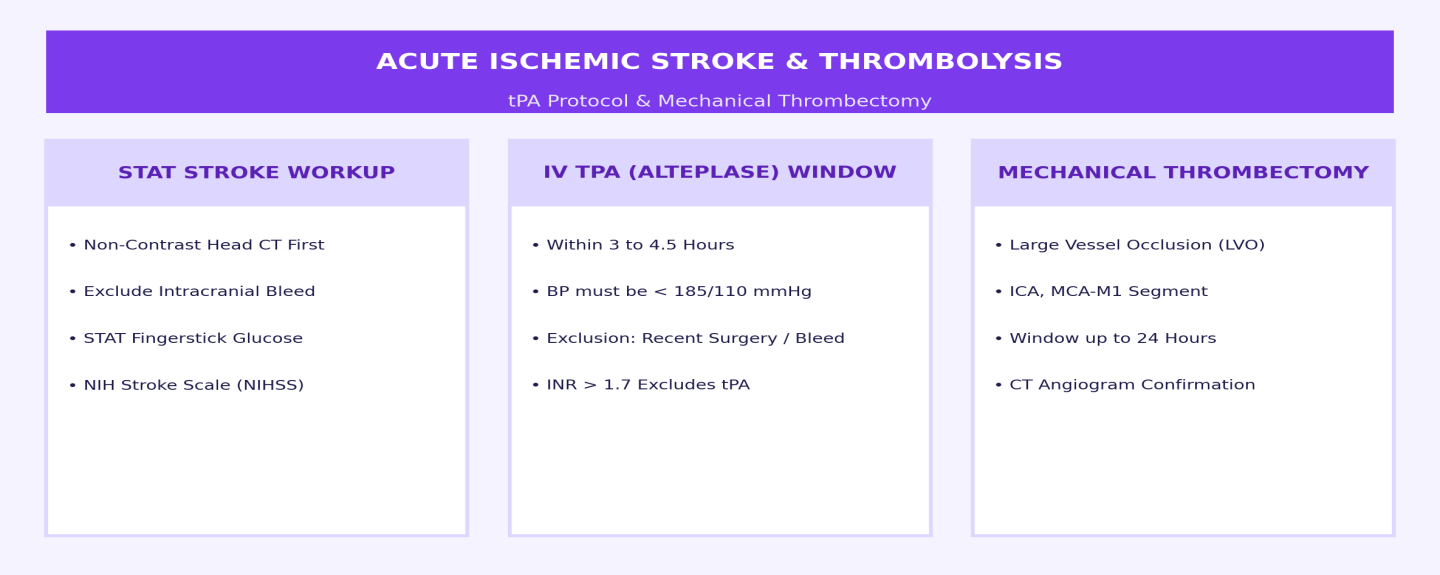


BOARD PEARL: Spinal Cord Injuries & Syndromes

Anterior Cord Syndrome = Loss of motor, pain, & temp BELOW lesion; Vibration & Proprioception are SPARED! Brown-Séquard = Ipsilateral motor loss, Contralateral pain/temp loss.

16. Acute Ischemic Stroke & Thrombolysis

Pathophysiology & Etiology: Acute focal neurologic deficit caused by thrombotic or embolic occlusion of a cerebral artery (MCA, ACA, PCA, Vertebrobasilar). Penumbra vs Core.
Clinical Presentation: MCA: Contralateral face/arm weakness > leg, sensory loss, aphasia (left) or neglect (right). ACA: Leg weakness > arm. Posterior: Ataxia, vertigo, diplopia.
Physical Examination: NIH Stroke Scale (NIHSS 0-42). Exclude stroke mimics (hypoglycemia, seizure, migraine).
Diagnostic Studies: STAT Non-Contrast Head CT (rule out bleed). Fingerstick Glucose. CTA Head/Neck (identify Large Vessel Occlusion - LVO).
Management & Treatment: IV tPA (Alteplase 0.9mg/kg) within 3 to 4.5 hours if BP < 185/110 mmHg and no contraindications. Mechanical Thrombectomy within 24 hours for LVO (ICA, MCA-M1).

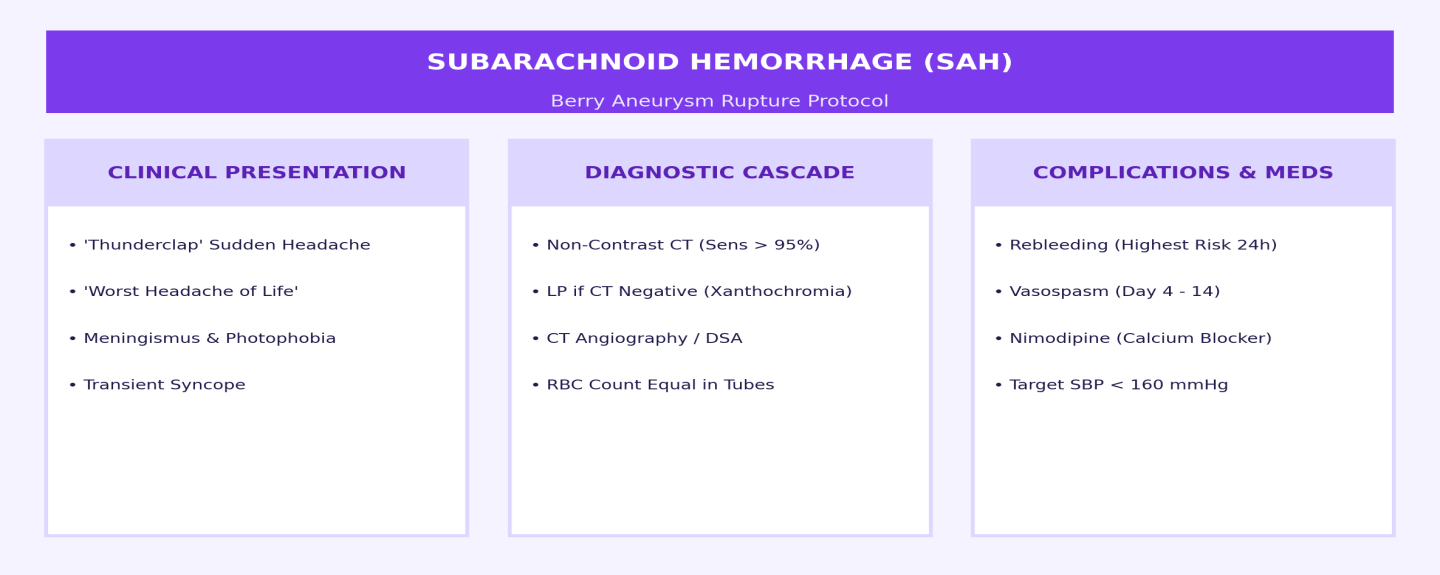


BOARD PEARL: Acute Ischemic Stroke & Thrombolysis

Time is Brain! Non-contrast CT Head first to exclude bleed. IV tPA window is 3 to 4.5 hours (BP must be < 185/110). Mechanical Thrombectomy up to 24 hours for Large Vessel Occlusion.

17. Subarachnoid Hemorrhage (SAH)

Pathophysiology & Etiology: Bleeding into the subarachnoid space, most commonly from rupture of a saccular (berry) aneurysm in the Circle of Willis (AComm > PComm > MCA). Trauma is most common overall.
Clinical Presentation: Sudden, explosive 'Thunderclap' headache peaking within seconds ('Worst headache of my life'). Nausea, vomiting, meningismus, transient LOC.
Physical Examination: Nuchal rigidity, Kernig/Brudzinski signs, subhyaloid retinal hemorrhages, CN III palsy (PComm aneurysm).
Diagnostic Studies: Non-Contrast Head CT (Sens > 95% in 6 hours). Lumbar Puncture MANDATORY if CT negative (looks for Xanthochromia - yellow CSF supernatant). CTA / Cerebral Angiography.
Management & Treatment: SBP control (< 160 mmHg w/ Nicardipine). Oral Nimodipine (60mg q4h) to prevent vasospasm. Emergent Neurosurgical coiling or clipping.

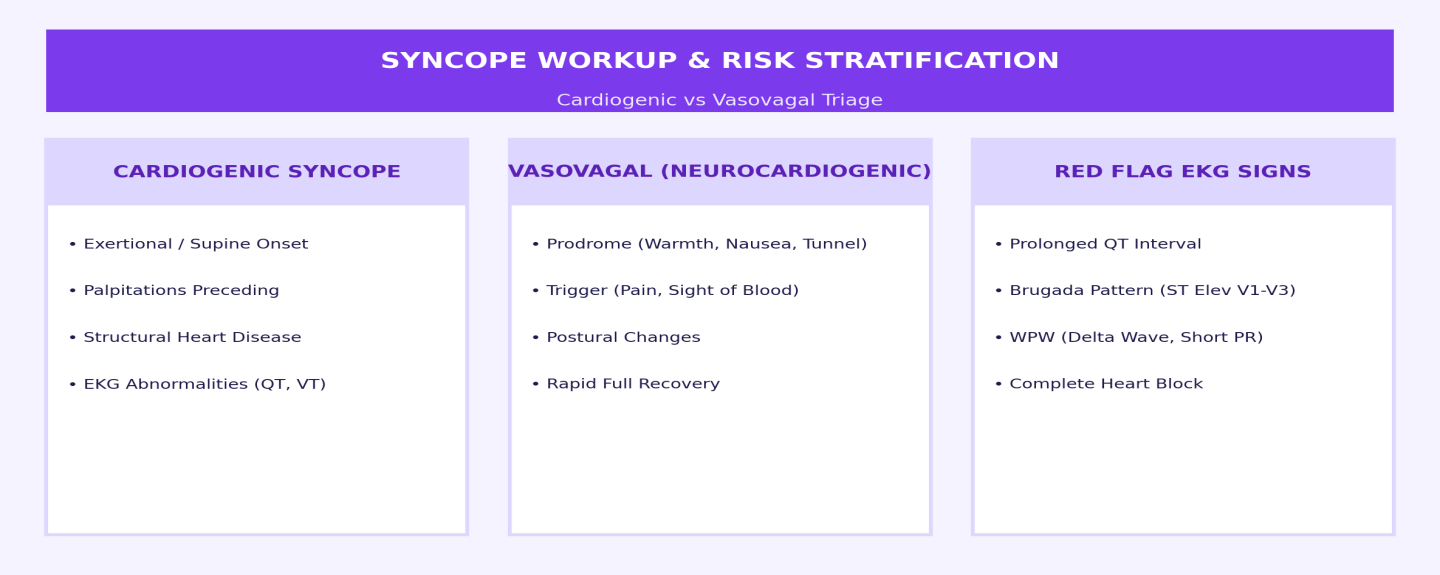


BOARD PEARL: Subarachnoid Hemorrhage (SAH)

If Non-Contrast CT is NEGATIVE but clinical suspicion for SAH remains high, a Lumbar Puncture is MANDATORY to test for Xanthochromia! Administer Nimodipine to prevent vasospasm.

18. Syncope Workup & Risk Stratification

Pathophysiology & Etiology: Transient loss of consciousness due to global cerebral hypoperfusion. Etiologies: Neurocardiogenic (Vasovagal - 50%), Orthostatic, Cardiac Arrhythmias, Structural Heart Disease.
Clinical Presentation: Vasovagal: Prodrome of dizziness, warmth, nausea, diaphoresis, tunnel vision. Cardiac: Sudden onset without prodrome, during exertion or supine.
Physical Examination: Orthostatic BP measurements (drop in SBP ≥ 20 or DBP ≥ 10), heart murmur (Aortic Stenosis, HOCM), neurological exam.
Diagnostic Studies: 12-Lead EKG for ALL patients (look for Brugada, Long QT, WPW, AV blocks, arrhythmogenic RV dysplasia). CBC, electrolytes, Echo if structural disease suspected.
Management & Treatment: Vasovagal: Reassurance, volume expansion. Cardiac: Admission, telemetry, pacemaker/ICD placement. High-Risk: San Francisco Syncope Rule.

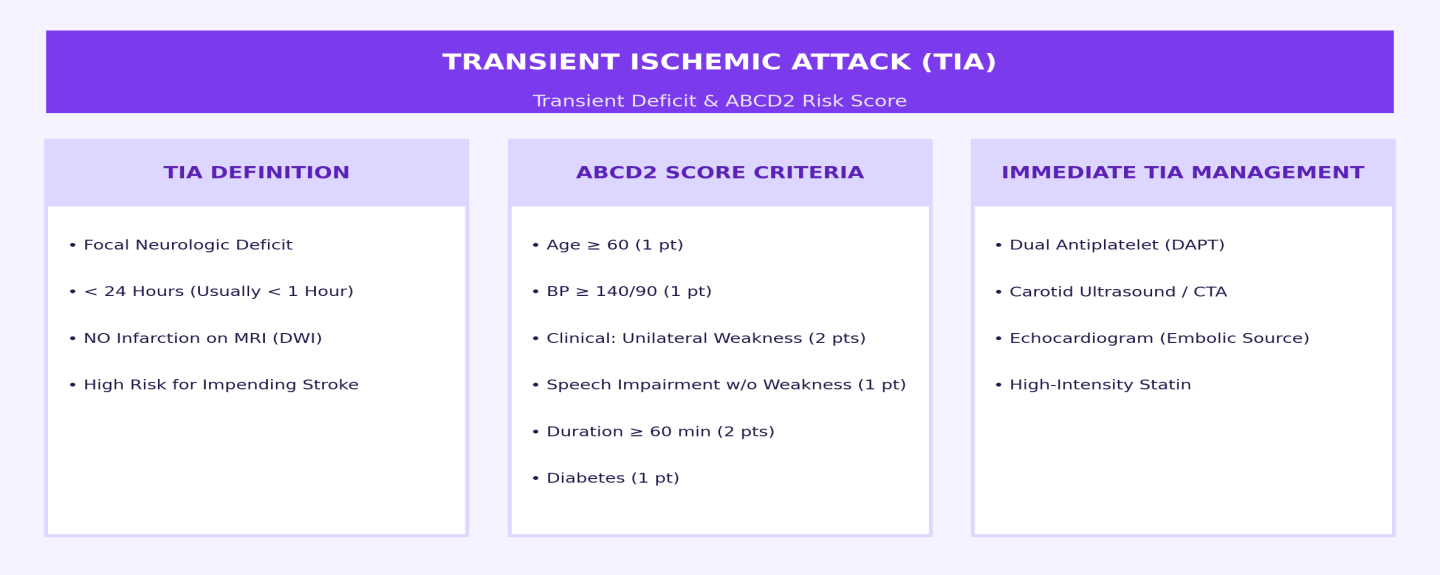


BOARD PEARL: Syncope Workup & Risk Stratification

Exertional syncope or syncope occurring while supine is CARDIAC until proven otherwise! Always order a 12-lead EKG to evaluate for fatal rhythm disorders.

19. Transient Ischemic Attack (TIA)

Pathophysiology & Etiology: Transient episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia, WITHOUT acute infarction on imaging (DWI MRI). High risk for subsequent stroke.
Clinical Presentation: Sudden focal weakness, numbness, speech difficulty (aphasia/dysarthria), or amaurosis fugax (painless monocular vision loss - 'curtain coming down'). Resolution within 1 hour.
Physical Examination: Carotid bruit, cardiac arrhythmia (Atrial Fibrillation), full neurologic assessment.
Diagnostic Studies: MRI Brain w/ DWI (gold standard). CTA Head/Neck or Carotid Ultrasound. 12-Lead EKG (rule out AFib). Echocardiogram. Calculate ABCD2 Score.
Management & Treatment: Dual Antiplatelet Therapy (DAPT: Aspirin + Clopidogrel x 21 days) for high-risk TIA (ABCD2 ≥ 4). High-intensity Statin (Atorvastatin 80mg). Carotid endarterectomy if > 70% stenosis.

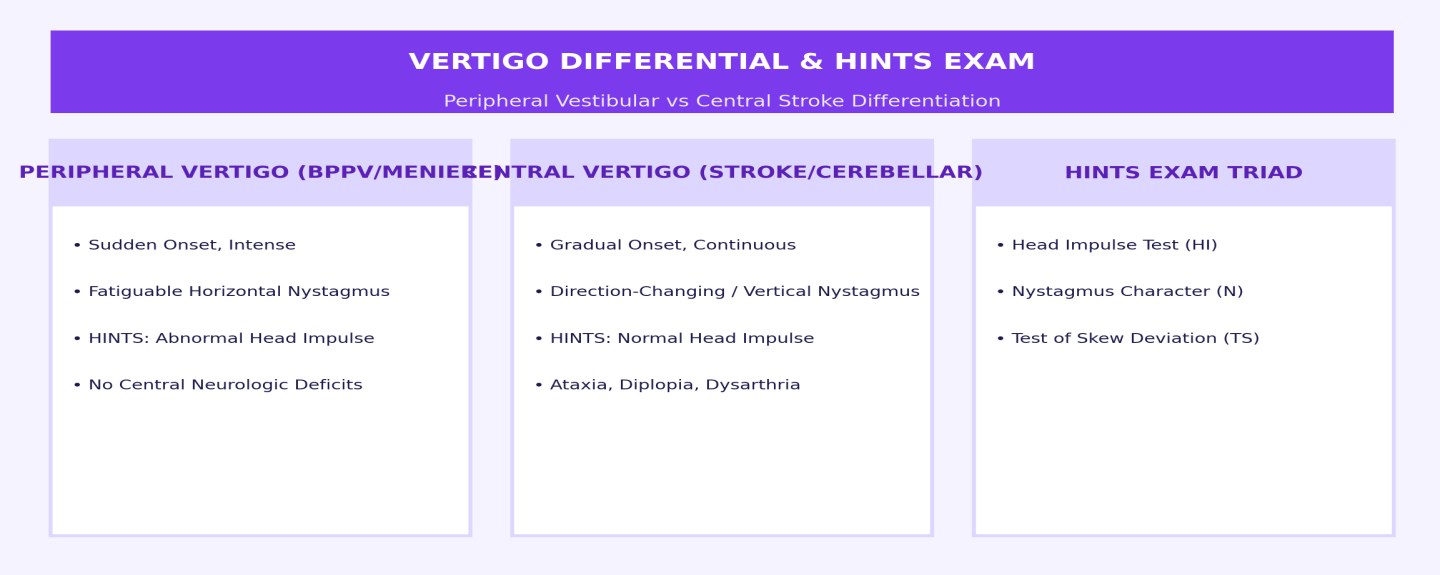


BOARD PEARL: Transient Ischemic Attack (TIA)

TIA is a major warning sign—highest stroke risk is within 48 hours! Calculate ABCD2 score and initiate Dual Antiplatelet Therapy (Aspirin + Clopidogrel) for high-risk patients.

20. Vertigo Differential & HINTS Exam

Pathophysiology & Etiology: Illusion of motion. Peripheral: Inner ear pathology (BPPV, Meniere's, Vestibular Neuritis). Central: Brainstem/Cerebellar stroke or bleed.
Clinical Presentation: Peripheral: Sudden, intense, positional, episodic, fatiguable horizontal nystagmus, hearing loss/tinnitus. Central: Constant, gradual, vertical/direction-changing nystagmus, ataxia.
Physical Examination: Perform **HINTS Exam** (Head Impulse, Nystagmus, Test of Skew). Central Stroke Sign: Normal Head Impulse, Direction-changing Nystagmus, or Vertical Skew Deviation.
Diagnostic Studies: HINTS exam is MORE sensitive than early MRI for stroke! Order STAT MRI/CT if HINTS suggests central cause or patient has severe ataxia.
Management & Treatment: BPPV: Epley Maneuver. Vestibular Neuritis: Meclizine, Benzo, Steroids. Central Vertigo: STAT Stroke Protocol.

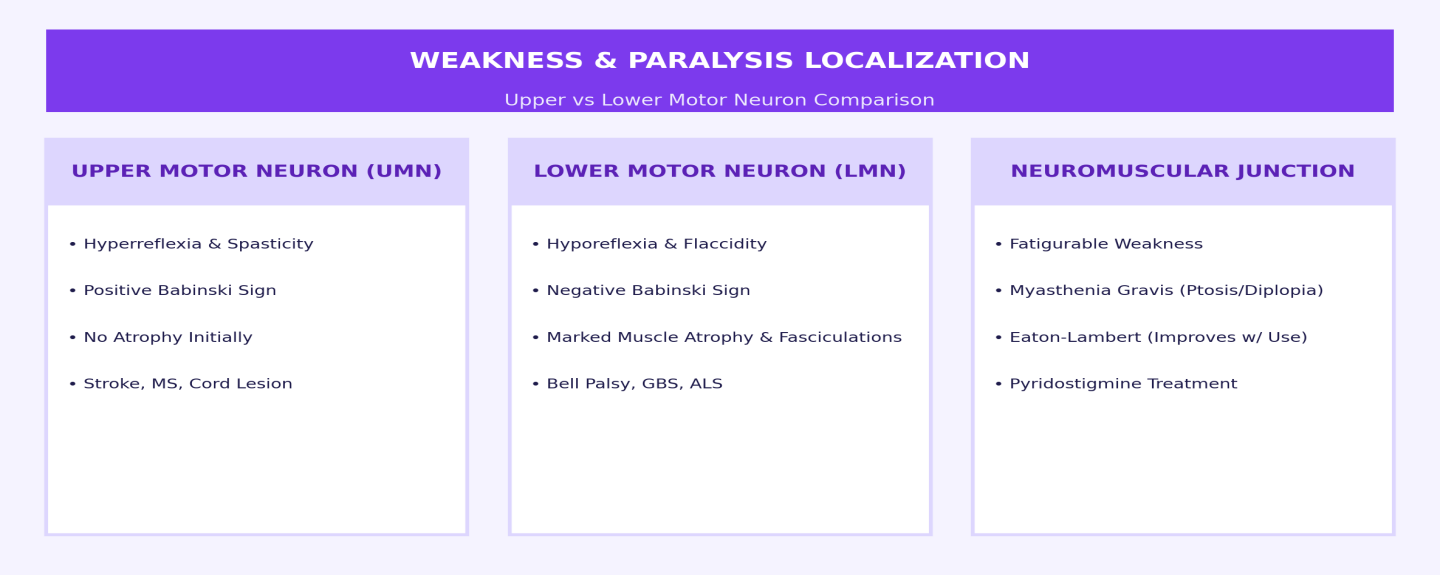


BOARD PEARL: Vertigo Differential & HINTS Exam

HINTS Exam for Acute Vestibular Syndrome: A NORMAL Head Impulse test in a patient with continuous vertigo strongly points to a CENTRAL STROKE, not a peripheral ear problem!

21. Weakness & Paralysis Localization

Pathophysiology & Etiology: Motor weakness localizes to Upper Motor Neuron (UMN - cortex, corticospinal tract), Lower Motor Neuron (LMN - anterior horn, nerve root, peripheral nerve), Neuromuscular Junction, or Muscle.
Clinical Presentation: UMN: Spastic paralysis, weakness. LMN: Flaccid paralysis, rapid atrophy. NMJ: Fatigurable weakness (Myasthenia Gravis).
Physical Examination: UMN vs LMN Matrix: UMN = Hyperreflexia, Increased Tone (Spasticity), Positive Babinski (Upgoing). LMN = Hyporeflexia/Areflexia, Decreased Tone (Flaccidity), Muscle Atrophy, Fasciculations, Negative Babinski.
Diagnostic Studies: MRI Brain/Spine for UMN/Radiculopathy. EMG/NCS for LMN/Neuropathy. Acetylcholine Receptor Antibodies & Ice Pack Test for Myasthenia Gravis.
Management & Treatment: Treat etiology. Myasthenia Gravis Crisis: Pyridostigmine, Plasmapheresis, IVIG. Avoid Succinylcholine in UMN/LMN lesions (severe hyperkalemia risk).



BOARD PEARL: Weakness & Paralysis Localization

UMN signs = Hyperreflexia, Spasticity, Upgoing Babinski (Stroke, MS, Cord Compression). LMN signs = Hyporeflexia, Flaccidity, Fasciculations, Muscle Atrophy (GBS, Bell's, ALS).